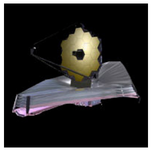

7396 - Exploring the Last Watering Hole of Low Metallicity PAH Emission: Deep Spectroscopic Observations of Sextans A

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Elizabeth Tarantino (PI)	**Space Telescope Science Institute**
Dr. Julia Christine Roman-Duval (CoI)	Space Telescope Science Institute
Dr. Martha L. Boyer (CoI)	Space Telescope Science Institute
Dr. Karl D. Gordon (CoI)	Space Telescope Science Institute
Dr. Bruce T. Draine (CoI)	Princeton University
Dr. Brandon S. Hensley (CoI)	California Institute of Technology
Dr. Christopher Clark (CoI) (ESA Member)	Space Telescope Science Institute - ESA - JWST
Dr. Karin Marie Sandstrom (CoI)	University of California - San Diego
Dr. Jeremy Chastenet (CoI) (ESA Member)	Ghent University
Dr. Benjamin F. Williams (CoI)	University of Washington
Dr. Kristen B W McQuinn (CoI)	Space Telescope Science Institute
Prof. JD Smith (CoI)	University of Toledo
Cory Whitcomb (CoI)	University of Toledo
Max J. B. Newman (CoI)	Space Telescope Science Institute
Dr. Alec S. Hirschauer (CoI)	Morgan State University
Dr. Logan H Jones (CoI)	Space Telescope Science Institute
Dr. Dries Van De Putte (CoI) (CSA Member)	The University of Western Ontario
Dr. Marjorie Decleir (CoI) (ESA Member)	Space Telescope Science Institute - ESA
Dr. Thomas Williams (CoI) (ESA Member)	University of Manchester
Dr. Ryan Chown (CoI)	Algoma University
Dr. Grace Telford (CoI)	University of Utah

Name	*Institution*
Dr. Thomas Lai (CoI)	California Institute of Technology
Dr. Adam Smercina (CoI)	Space Telescope Science Institute

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
SexA				
	1	MIRI MRS-A	MIRI Medium Resolution Spectroscopy	(1) SexA_PAH-MRS
	2	MIRI MRS-B	MIRI Medium Resolution Spectroscopy	(1) SexA_PAH-MRS
	11	MIRI MRS-B	MIRI Medium Resolution Spectroscopy	(1) SexA_PAH-MRS
	3	MIRI MRS-C	MIRI Medium Resolution Spectroscopy	(1) SexA_PAH-MRS
	4	MIRI MRS Background-A	MIRI Medium Resolution Spectroscopy	(2) SexA_PAH-MRS_BKGR
	5	MIRI MRS Background-B	MIRI Medium Resolution Spectroscopy	(2) SexA_PAH-MRS_BKGR
	12	MIRI MRS Background-B	MIRI Medium Resolution Spectroscopy	(2) SexA_PAH-MRS_BKGR
	6	MIRI MRS Background-C	MIRI Medium Resolution Spectroscopy	(2) SexA_PAH-MRS_BKGR
	7	NIRSpec	NIRSpec IFU Spectroscopy	(3) SexA_PAH-NIRSpec
	8	NIRSpec Background	NIRSpec IFU Spectroscopy	(4) SexA_PAH-NIRSpec_BKGR
	9	MIRI LRS	MIRI Low Resolution Spectroscopy	(8) SexA_PAH-LRS
	10	MIRI LRS Background	MIRI Low Resolution Spectroscopy	(9) SexA_PAH-LRS_BKGR

ABSTRACT

In typical star-forming galaxies, the mid-infrared (MIR) spectrum is dominated by emission from Polycyclic Aromatic Hydrocarbons (PAHs), the smallest dust grains found in the interstellar medium (ISM). In regions with low metallicities, the fraction of PAHs in the ISM undergoes a significant decline. Recent JWST photometric observations of the dwarf galaxy Sextans A, however, reveal a remarkable detection of PAHs in a 7% solar metallicity environment, making it the lowest metallicty detection of PAHs to date. We propose deep MIRI MRS and NIRSpec IFU observations on the brightest PAH emitting clump found through imaging. With full spectral coverage across the 3-18 micron range, we will identify PAH features that were impossible to see through photometry alone, determine the low metallicity PAH properties through band ratios, constrain the molecular gas environment near the PAHs with the H2 rotational transitions accessible only through spectroscopy, and create the low metallicity MIR spectroscopic template for future JWST observations and models. The low metallicity MIR spectrum combined with ancillary data (HST,

MUSE, ALMA, and VLA) will uniquely constrain the lifecycle of the PAHs, which is vital for understanding their importance at high redshift where metallicities are much lower than solar. Sextans A is the ideal galaxy to provide these constraints, as it is the only galaxy with metallicity less than 10% solar that has a robust detection of PAHs and its proximity allows for a detailed, resolved study of PAHs. JWST is the only telescope suitable for these observations due to its superior sensitivity, high spatial resolution, and coverage of the MIR wavelengths.

OBSERVING DESCRIPTION

MIRI MRS

We will observe the bright PAH-emitting region in Sextans A with full spectral coverage of the MRS. Due to the data volume limits, we use the SLOWR1 readout for MRS and 20 groups per integration (400 s integrations) to limit cosmic ray contamination. We request 8 integrations per exposure and use the 4-point dither optimized for extended sources, leading to an exposure time of 4.5 hours for each of the three spectral settings. We request dedicated background observations using identical detector settings as is recommended in the documentation for faint, extended sources. The long integration time (> 24 hours) prohibit the background to be scheduled uninterruptible with the science observations. Instead, we split each MRS observation into separate SHORT (A), MEDIUM (B), and LONG (C) visits that are each grouped with an associated background observation in order to facilitate schedulability. These visits are grouped within seven days of each other to ensure negligible difference in the position angle (PA).

We will use the MRS simultaneous MIRI imaging mode to acquire observations of the F2100W filter during the MRS dedicated background ("off") observations by setting the primary channel as the IMAGER, as is recommended in the documentation. To ensure the sky background does not saturate with the F2100W integration, we choose the FASTR1 readout pattern and use 29 groups per integration, 40 integrations per group, and 4 dithers, leading to an exposure time of 3.8 hours. The simultaneous imaging for MRS "on" is set to the F560W, F770W, and F1000W filters to improve the astrometry of the MRS field.

NIRSpec IFU

As recommended in the JWST documentation, we use the NRSIRS2 readout mode which is optimized for long exposure IFU observations of faint targets and minimizes data volume. We request 15 groups per integration, 2 integrations, and use the 4-POINT-DITHER for extended sources for a total of 2.5 hours. To minimize the contribution of leakage from the MSA, we will acquire a leakcal exposure for the first dither. Due to the faint and extended nature of the PAH emission, we request a dedicated background observation that is non-interruptable with the science observations and uses identical detector settings.

Target Acquisition

The MRS and NIRSpec field of view will cover the faint, extended PAH emission and are not suitable for TAQ. Instead, existing MIRI and NIRCam imaging is used to find offset TAQ sources. We identify a source within 3" of the NIRSpec field of view that provides an SNR 30 in the F110W filter. For MRS, we find a bright star 8" away from the MRS pointing (within the recommended visit splitting maximum of 15") that achieves an SNR 30 using the F560W filter.

Special Requirements

We request that the MRS observations be taken in a low background as the SNR changes >25% between high and low backgrounds. We also require each visit to be uninterruptible with its associated background and group all visits to occur with 7 days to minimize the impact of PA changes. For NIRSpec, we request that the background and science observations are non-interruptable and require the PA to be within 75--125 to ensure the MSA is not on a bright source.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SexA_PAH-MRS	RA: 10 11 6.8875 (152.7786979d) Dec: -04 42 17.64 (-4.70490d) Equinox: J2000	Epoch of Position: 2023	
Comments: Category=ISM Description=[Dense interstellar clouds, Interstellar clouds, Interstellar dust, Interstellar molecules] Extended=YES				
(2)	SexA_PAH-MRS_BKGR	RA: 10 11 4.8116 (152.7700483d) Dec: -04 42 19.63 (-4.70545d) Equinox: J2000	Epoch of Position: 2023	
Comments: Category=Unidentified Description=[Blank field] Extended=YES				
(3)	SexA_PAH-NIRSpec	RA: 10 11 6.8875 (152.7786979d) Dec: -04 42 17.64 (-4.70490d) Equinox: J2000	Epoch of Position: 2023	
Comments: Category=ISM Description=[Dense interstellar clouds, Interstellar clouds, Interstellar dust, Interstellar molecules] Extended=YES				
(4)	SexA_PAH-NIRSpec_BKGR	RA: 10 11 7.9629 (152.7831788d) Dec: -04 42 49.95 (-4.71387d) Equinox: J2000	Epoch of Position: 2023	
Comments: Category=Unidentified Description=[Blank field] Extended=YES				
(7)	VPW98-298	RA: 10 11 6.3799 (152.7765829d) Dec: -04 42 18.16 (-4.70504d) Equinox: J2000	Proper Motion RA: 0.784 mas/yr Proper Motion Dec: 0.788 mas/yr Parallax: 1.077E-4" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[F supergiants, G supergiants] Extended=NO				
(8)	SexA_PAH-LRS	RA: 10 11 6.9214 (152.7788392d) Dec: -04 42 18.10 (-4.70503d) Equinox: J2000	Epoch of Position: 2023	
Comments: Category=ISM Description=[Dense interstellar clouds, Interstellar clouds, Interstellar dust, Interstellar molecules] Extended=YES				

	(9)	SexA_PAH-LRS_BKGR	RA: 10 11 7.8287 (152.7826196d) Dec: -04 42 23.54 (-4.70654d) Equinox: J2000	Epoch of Position: 2023
	Comments: *Category=Unidentified* *Description=[Blank field]* *Extended=YES*			

Observation

Proposal 7396, Observation 1: MIRI MRS-A

Tue Jun 02 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[MIRI MRS Background-A (Obs 4)]

Diagnostics

(MIRI MRS-A (Obs 1)) Warning (Form): Filter mismatch between science and background observations may result in incorrect background subtraction.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SexA_PAH-MRS	RA: 10 11 6.8875 (152.7786979d) Dec: -04 42 17.64 (-4.70490d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=ISM
Description=[Dense interstellar clouds, Interstellar clouds, Interstellar dust, Interstellar molecules]
Extended=YES

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	7 VPW98-298	F560W	FASTGRPAVG	4	1	1	44.401	289548

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F560W	FASTR1	90	14	1	Dither 1	4	56	14130.504	220052
1	SHORT(A)	MRSLONG		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052
1	SHORT(A)	MRSSHORT		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052

Special Requirements	
	Background Limited. Background no more than 30th percentile above minimum Group Observations 1, 4, Non-interruptible Group Observations 1, 2, 3, 4, 5, 6 within 7 Days

Observation

Proposal 7396, Observation 2: MIRI MRS-B

Tue Jun 02 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[MIRI MRS Background-B (Obs 5)]

Diagnostics

(MIRI MRS-B (Obs 2)) Warning (Form): Filter mismatch between science and background observations may result in incorrect background subtraction.

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SexA_PAH-MRS	RA: 10 11 6.8875 (152.7786979d) Dec: -04 42 17.64 (-4.70490d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=ISM
Description=[Dense interstellar clouds, Interstellar clouds, Interstellar dust, Interstellar molecules]
Extended=YES

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	7 VPW98-298	F560W	FASTGRPAVG	4	1	1	44.401	289548

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	90	14	1	Dither 1	4	56	14130.504	220052
1	MEDIUM(B)	MRSLONG		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052
1	MEDIUM(B)	MRSSHORT		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052

Special Requirements	Background Limited. Background no more than 30th percentile above minimum Group Observations 2, 5, Non-interruptible Group Observations 1, 2, 3, 4, 5, 6 within 7 Days

Observation

Proposal 7396, Observation 11: MIRI MRS-B

Tue Jun 02 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[MIRI MRS Background-B (Obs 12)]

Diagnostics

(MIRI MRS-B (Obs 11)) Warning (Form): Filter mismatch between science and background observations may result in incorrect background subtraction.

(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SexA_PAH-MRS	RA: 10 11 6.8875 (152.7786979d) Dec: -04 42 17.64 (-4.70490d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=ISM
Description=[Dense interstellar clouds, Interstellar clouds, Interstellar dust, Interstellar molecules]
Extended=YES

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	7 VPW98-298	F560W	FASTGRPAVG	4	1	1	44.401	289548

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	90	14	1	Dither 1	4	56	14130.504	220052
1	MEDIUM(B)	MRSLONG		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052
1	MEDIUM(B)	MRSSHORT		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052

Special Requirements	
	Background Limited. Background no more than 30th percentile above minimum Sequence Observations 12, 11 (reordered), Non-interruptible

Observation

Proposal 7396, Observation 3: MIRI MRS-C

Tue Jun 02 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[MIRI MRS Background-C (Obs 6)]

Diagnostics

(MIRI MRS-C (Obs 3)) Warning (Form): Filter mismatch between science and background observations may result in incorrect background subtraction.

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	SexA_PAH-MRS	RA: 10 11 6.8875 (152.7786979d) Dec: -04 42 17.64 (-4.70490d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=ISM
Description=[Dense interstellar clouds, Interstellar clouds, Interstellar dust, Interstellar molecules]
Extended=YES

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	7 VPW98-298	F560W	FASTGRPAVG	4	1	1	44.401	289548

Template

Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F1000W	FASTR1	90	14	1	Dither 1	4	56	14130.504	220052
1	LONG(C)	MRSLONG		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052
1	LONG(C)	MRSSHORT		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052

Special Requirements	
	Background Limited. Background no more than 30th percentile above minimum Group Observations 3, 6, Non-interruptible Group Observations 1, 2, 3, 4, 5, 6 within 7 Days

Observation

Proposal 7396, Observation 4: MIRI MRS Background-A

Tue Jun 02 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [MIRI MRS-A (Obs 1)]

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SexA_PAH-MRS_BKGR	RA: 10 11 4.8116 (152.7700483d) Dec: -04 42 19.63 (-4.70545d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=Unidentified
Description=[Blank field]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	Imager	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F2100W	FASTR1	29	40	1	Dither 1	4	160	13309.092	220052
1	SHORT(A)	MRSLONG		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052
1	SHORT(A)	MRSSHORT		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052

Special Requirements	
	Aperture PA Range 102.83544897 to 107.83544897 Degrees (V3 98.0 to 103.0) Background Limited. Background no more than 30th percentile above minimum Group Observations 1, 4, Non-interruptible Group Observations 1, 2, 3, 4, 5, 6 within 7 Days

Observation

Proposal 7396, Observation 5: MIRI MRS Background-B

Tue Jun 02 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [MIRI MRS-B (Obs 2)]

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SexA_PAH-MRS_BKGR	RA: 10 11 4.8116 (152.7700483d) Dec: -04 42 19.63 (-4.70545d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=Unidentified
Description=[Blank field]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	Imager	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F2100W	FASTR1	29	40	1	Dither 1	4	160	13309.092	220052
1	MEDIUM(B)	MRSLONG		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052
1	MEDIUM(B)	MRSSHORT		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052

Special Requirements	
	Aperture PA Range 102.83544897 to 107.83544897 Degrees (V3 98.0 to 103.0) Background Limited. Background no more than 30th percentile above minimum Group Observations 2, 5, Non-interruptible Group Observations 1, 2, 3, 4, 5, 6 within 7 Days

Observation

Proposal 7396, Observation 12: MIRI MRS Background-B

Tue Jun 02 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [MIRI MRS-B (Obs 11)]

Diagnostics

(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SexA_PAH-MRS_BKGR	RA: 10 11 4.8116 (152.7700483d) Dec: -04 42 19.63 (-4.70545d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=Unidentified
Description=[Blank field]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	Imager	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F2100W	FASTR1	29	40	1	Dither 1	4	160	13309.092	220052
1	MEDIUM(B)	MRSLONG		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052
1	MEDIUM(B)	MRSSHORT		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052

Special Requirements	
	Aperture PA Range 102.83544897 to 107.83544897 Degrees (V3 98.0 to 103.0) Background Limited. Background no more than 30th percentile above minimum Sequence Observations 12, 11 (reordered), Non-interruptible

Observation

Proposal 7396, Observation 6: MIRI MRS Background-C

Tue Jun 02 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [MIRI MRS-C (Obs 3)]

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SexA_PAH-MRS_BKGR	RA: 10 11 4.8116 (152.7700483d) Dec: -04 42 19.63 (-4.70545d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=Unidentified
Description=[Blank field]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	Imager	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F2100W	FASTR1	29	40	1	Dither 1	4	160	13309.092	220052
1	LONG(C)	MRSLONG		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052
1	LONG(C)	MRSSHORT		SLOWR1	21	7	1	Dither 1	4	28	14620.631	220052

Special Requirements	
	Aperture PA Range 102.83544897 to 107.83544897 Degrees (V3 98.0 to 103.0) Background Limited. Background no more than 30th percentile above minimum Group Observations 3, 6, Non-interruptible Group Observations 1, 2, 3, 4, 5, 6 within 7 Days

Observation

Proposal 7396, Observation 7: NIRSpec

Tue Jun 02 17:00:12 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observations:[NIRSpec Background (Obs 8)]

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	SexA_PAH-NIRSpec	RA: 10 11 6.8875 (152.7786979d) Dec: -04 42 17.64 (-4.70490d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=ISM
Description=[Dense interstellar clouds, Interstellar clouds, Interstellar dust, Interstellar molecules]
Extended=YES

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	7 VPW98-298	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	289548

Template

HFF Readout Mode
false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2	15	2	false	true	NONE	4	8	8870.045	220052
2	G395M/F290LP	NRSIRS2	15	2	true	false	NONE	1	2	2217.511	220052

Special Requirements	
	Aperture PA Range 213.97164917 to 263.97164917 Degrees (V3 75.0 to 125.0) Group Observations 7, 8, Non-interruptible

Observation

Proposal 7396, Observation 8: NIRSpec Background

Tue Jun 02 17:00:13 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observation For: [NIRSpec (Obs 7)]

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	SexA_PAH-NIRSpec_BKGR	RA: 10 11 7.9629 (152.7831788d) Dec: -04 42 49.95 (-4.71387d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=Unidentified
Description=[Blank field]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2	15	2	false	true	NONE	4	8	8870.045	220052

Special Requirements

Aperture PA Range 213.97164917 to 263.97164917 Degrees (V3 75.0 to 125.0)

Group Observations 7, 8, Non-interruptible

Observation

Proposal 7396, Observation 9: MIRI LRS

Tue Jun 02 17:00:13 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Diagnostics

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(8)	SexA_PAH-LRS	RA: 10 11 6.9214 (152.7788392d) Dec: -04 42 18.10 (-4.70503d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=ISM
Description=[Dense interstellar clouds, Interstellar clouds, Interstellar dust, Interstellar molecules]
Extended=YES

Acquisition

#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	7 VPW98-298	F560W	FASTGRPAVG	4	1	1	44.401	289548

Template

Subarray	Obtain Verification Image?
FULL	true

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Pointing Verification

#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	Optional ETC ID	Filter
1	FASTR1	100	2	2	1	1	557.783		F770W

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
1	FASTR1	75	9	18	1	2	3790.705	

Special Requirements

Background Limited. Background no more than 30th percentile above minimum

Sequence Observations 9, 10, Non-interruptible

Observation

Proposal 7396, Observation 10: MIRI LRS Background

Tue Jun 02 17:00:13 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Low Resolution Spectroscopy

Diagnostics

(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(9)	SexA_PAH-LRS_BKGR	RA: 10 11 7.8287 (152.7826196d) Dec: -04 42 23.54 (-4.70654d) Equinox: J2000	Epoch of Position: 2023	

Comments:
Category=Unidentified
Description=[Blank field]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Subarray	Obtain Verification Image?
F560W	FULL	false

Dithers

#	Dither Type	No. Spectral Steps	Spectral Step Offset	No. Spatial Steps	Spatial Step Offset
1	ALONG SLIT NOD				

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	Optional ETC ID
1	FASTR1	75	9	18	1	2	3790.705	

Special Requirements	
	Background Limited. Background no more than 30th percentile above minimum Sequence Observations 9, 10, Non-interruptible